

IIT Guwahati - Department of Computer Science & Engineering

CS 223- Computer Organization & Architecture –Quiz 2 (06.03.2021)

1. Consider a 5-stage in-order RISC instruction pipeline. Consider a program that has a sequence of instructions of the form LOAD, ADD, STORE, SUB that is repeated 100 times in that order. So the program has a total of 400 instructions. The ADD has a data dependency on the value accessed from memory by the immediate LOAD before it. The STORE and SUB have a data dependency on the resultant of the ADD instruction immediately preceding them. Every LOAD instruction (except the first instruction) has a data dependency on the resultant of immediate SUB before it. Full operand forwarding is permitted to reduce data hazards. Assume that the IF for the first LOAD is done at clock cycle number 101.
 - (a) What is the 150th instruction?
 - (b) In which clock cycle number, the 60th LOAD performs MEM stage?
 - (c) What is the total number of stalls incurred during this program?
 - (d) What is the average CPI (cycles/instruction) for this program?

Ans:

I1: LOAD

I2: ADD

I3: STORE

I4: SUB

I2 depends on I1

I3 and I4 depend on I2

LOAD depends on SUB except the first instruction

(a) ADD

148th instruction is SUB , 149 is LOAD and 150 is ADD

(b) 60th Load performs MEM stage at cycle

Iteration	Instruction	101	102	103	104	105	106	107	108	109	110	111	112	113	114
1	Load	F	D	E	M	W									
	Add		F	Stall	D	E	M	W							
	Store				F	D	E	M	W						
	Sub					F	D	E	M	W					
2	Load						F	D	E	M	W				
	Add							F	Stall	D	E	M	W		
	Store									F	D	E	M	W	
	Sub										F	D	E	M	W

$$= 104 + (60-1) * 5$$

$$= 399$$

(c) Only one stall in each iteration , so total number stalls = 100

(d) Average CPI = Total Number of cycles required for execution /
Total Number of Instructions

$$= \text{Time required for } 100^{\text{th}} \text{ SUB} / 4 \times 100$$

$$= 109 + (99 \times 5) - 100 / 400$$

$$= 1.26$$

2. Consider the following MIPS code. Assume the IF stage for the first L.D instruction happens at clock cycle 1.

(a) In which clock cycle the S.D instruction writes the result to memory?

(b) Will compiler scheduling reduce the number of stalls? Why/Why not?

```

L.D      F2, 0(R1)
MUL.D    F4, F2, F0
L.D      F6, 0(R2)
ADD.D    F6, F4, F6
S.D      F6, 0(R2)
  
```

Instruction	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Load	F	D	E	M	W																		
Mul		F	D	stall	E1	E2	E3	E4	E5	E6	E7	M	W										
Load			F	stall	D	E	M	W															
Add						F	D	stall	stall	stall	stall	E1	E2	E3	E4	M	W						
Store							F	stall	stall	stall	stall	D	stall	stall	stall	E	M	W					

(a) 18th Cycle

(b) Can execute LOAD first, but will not reduce the number of stalls

Instruction	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Load	F	D	E	M	W																		
Load		F	D	E	M	W																	
Mul			F	D	E1	E2	E3	E4	E5	E6	E7	M	W										
Add				F	D	stall	stall	stall	stall	stall	stall	E1	E2	E3	E4	M	W						
Store						F	stall	stall	stall	stall	stall	D	stall	stall	stall	E	M	W					

3. Consider the following MIPS code snippet that is part of a large program. What are the kind of hazards possible here? How can you reduce/eliminate them?

```

MUL.D    F1, F2, F3
ADD.D    F4, F5, F6
ADD.D    F7, F4, F8
L.D      F4, 8(R0)
ADD.D    F8, F4, F1
  
```

RAW hazard on F4 between ADD.D F4, F5, F6 and ADD.D F7, F4, F8.

WAR hazard on F8 between ADD.D F7, F4, F8 and ADD.D F8, F4, F1.
It can be eliminated using operand forwarding with less number of cycles.

4. In a program there is a branch instruction which is iterated 6 times. The processor uses a (2, 2) correlating branch predictor. Each entry of BHT is a 2-bit value that is updated by a standard 2-bit, 4-state finite state machine. Let the initial entry of the BHT for NN/NT/TN/TT is 00/00/11/11, respectively. It was found that all the 6 branch instances are mis-predicted. The BHT is indexed with an NN value initially. Illustrate the operation of the branch predictor with the help of a neat labelled BHT.
- (a) List the actual outcomes of the 6 iterations of this branch instruction in order.
- (b) What is the BHT entry after knowing the outcome of the 6th iteration of this branch instruction?

S/N	Previous outcome	BHT NN/NT/TN/TT	Prediction	Outcome
1.	NN	00/00/11/11	N	T
2.	NT	01/00/11/11	N	T
3.	TT	01/01/11/11	T	N
4.	TN	01/01/11/10	T	N
5.	NN	01/01/10/10	N	T
6.	NT	11/01/10/10	N	T
7.	TT	11/11/10/10	—	—

[4 MARKS]

- a) Actual outcomes – T T N N T T [1 MARK]
- b) BHT entry after 6th iteration – 11/11/10/10 [1 MARK]

5. In a program there is a branch instruction which is iterated 6 times. The processor uses a (2, 2) correlating branch predictor. Each entry of BHT is a 2-bit value that is updated by a standard 2-bit, 4-state finite state machine. Let the initial entry of the BHT for NN/NT/TN/TT is 00/00/11/11, respectively. It was found that out of the 6 branch instances only the last 3 are mis-predicted. The BHT is indexed with an NN value initially. Illustrate the operation of the branch predictor with the help of a neat labelled BHT.
- (a) List the actual outcomes of the 6 iterations of this branch instruction in order.
- (b) What is the BHT entry after knowing the outcome of the 6th iteration of this branch instruction?

Sol.

S/N	Previous outcome	BHT NN/NT/TN/TT	Prediction	Outcome
1.	NN	00/00/11/11	N	N
2.	NN	00/00/11/11	N	N
3.	NN	00/00/11/11	N	N
4.	NN	00/00/11/11	N	T
5.	NT	01/00/11/11	N	T
6.	TT	01/01/11/11	T	N
7.	TN	01/01/11/10	—	—

[4 MARKS]

- a) Actual outcomes – N N N T T N [1 MARK]
b) BHT entry after 6th iteration – 01/01/11/10 [1 MARK]

6. In a program there is a branch instruction which is iterated 6 times. The processor uses a (2, 2) correlating branch predictor. Each entry of BHT is a 2-bit value that is updated by a standard 2-bit, 4-state finite state machine. Let the initial entry of the BHT for NN/NT/TN/TT is 00/00/11/11, respectively. At the end of prediction for 6 instances of branches and subsequent state update after branch outcome, the BHT for NN/NT/TN/TT is 01/01/10/10, respectively. The branch predictor has an accuracy of 0.333. The BHT is indexed with an NN value initially. Illustrate the operation of the branch predictor with the help of a neat labelled BHT.

- (a) List the actual outcomes of the 6 iterations of this branch instruction in order.
(b) How many times the predictor changes its state?

Sol. There are multiple answers possible for this question

(c) 4 times

S/N	Previous outcome	BHT NN/NT/TN/TT	Prediction	Outcome
1.	NN	00/00/11/11	N	T
2.	NT	01/00/11/11	N	T
3.	TT	01/01/11/11	T	T
4.	TT	01/01/11/11	T	T
5.	TT	01/01/11/11	T	N
6.	TN	01/01/11/10	T	N

7.	NN	01/01/10/10	—	—
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S/N	Previous outcome	BHT NN/NT/TN/TT	Prediction	Outcome
1.	NN	00/00/11/11	N	N
2.	NN	00/00/11/11	N	N
3.	NN	00/00/11/11	N	T
4.	NT	01/00/11/11	N	T
5.	TT	01/01/11/11	T	N
6.	TN	01/01/11/10	T	N
7.	NN	01/01/10/10	—	—

S/N	Previous outcome	BHT NN/NT/TN/TT	Prediction	Outcome
1.	NN	00/00/11/11	N	N
2.	NN	00/00/11/11	N	N
3.	NN	00/00/11/11	N	T
4.	NT	01/00/11/11	N	T
5.	TT	01/01/11/11	T	N
6.	TN	01/01/11/10	T	N
7.	NN	01/01/10/10	—	—

S/N	Previous outcome	BHT NN/NT/TN/TT	Prediction	Outcome
1.	NN	00/00/11/11	N	N
2.	NN	00/00/11/11	N	T
3.	NT	01/00/11/11	N	T
4.	TT	01/01/11/11	T	T

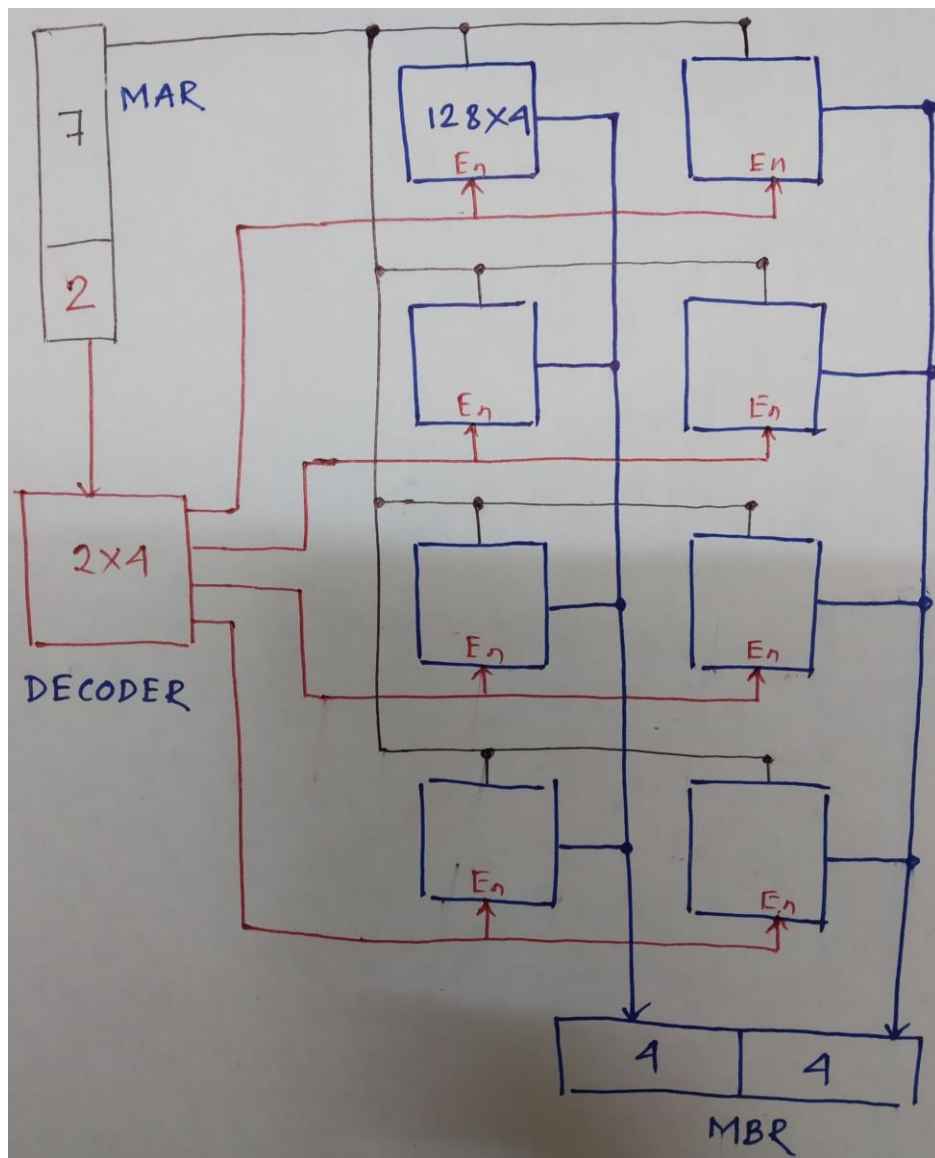
5.	TT	01/01/11/11	T	N
6.	TN	01/01/11/10	T	N
7.	NN	01/01/10/10	—	—

7. To design a 512 X 8 memory using 128 X 4 chips:

- How many chips will be required? (1-mark)
- How many rows and columns will be there? (1-mark)
- Show the design layout with all the necessary units including decoder, MAR and MBR. (3-marks)

Solution:

- Chips: $512 \times 8 / 128 \times 4 = 8$
- Rows: $512 / 128 = 4$, Columns: $8 / 4 = 2$
-

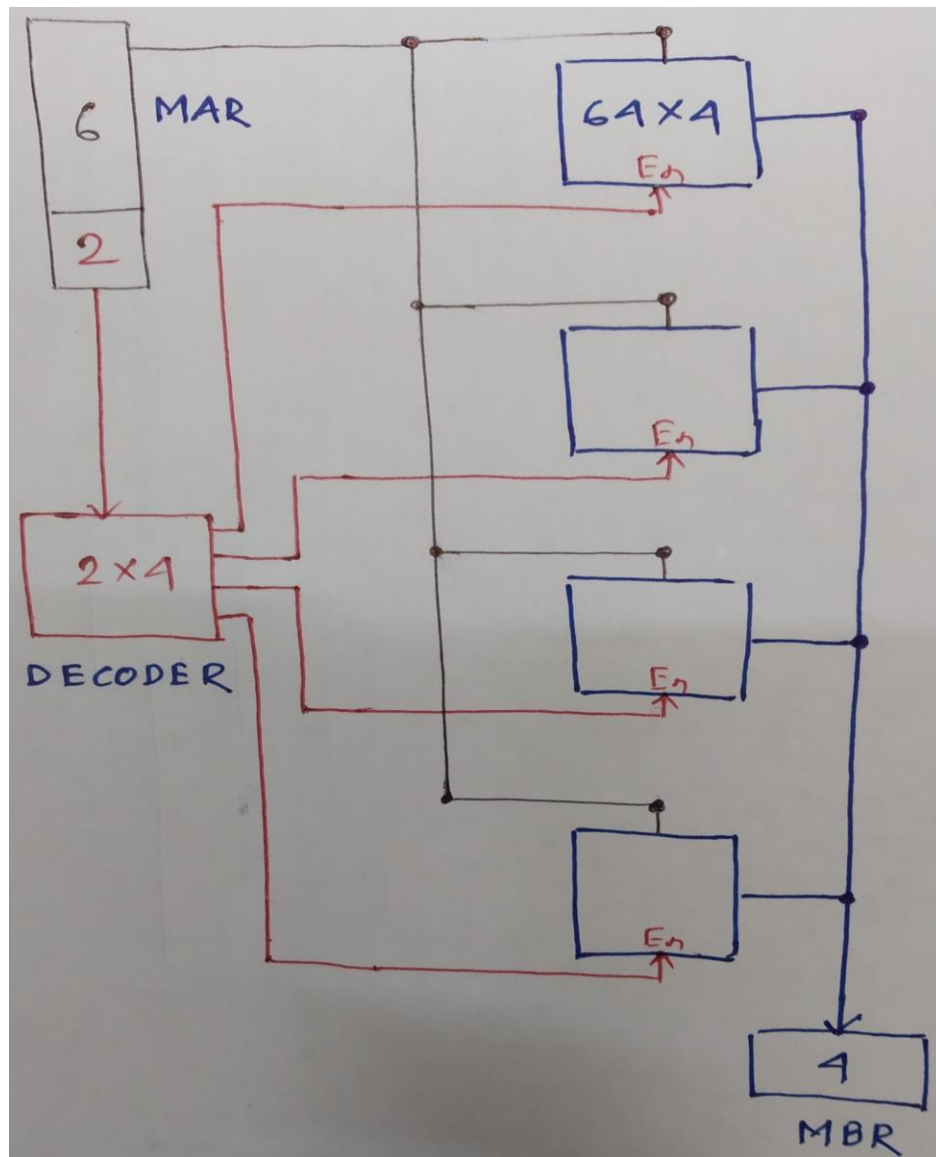


8. To design a 256 X 4 memory using 64 X 4 chips:
- How many chips will be required? (1-mark)
 - How many rows and columns will be there? (1-mark)
 - Show the design layout with all the necessary units including decoder, MAR and MBR. (3-marks)

Solution:

Chips: $256 \times 4 / 64 \times 4 = 4$

Rows: $256 / 64 = 4$, **Columns:** $4 / 4 = 1$

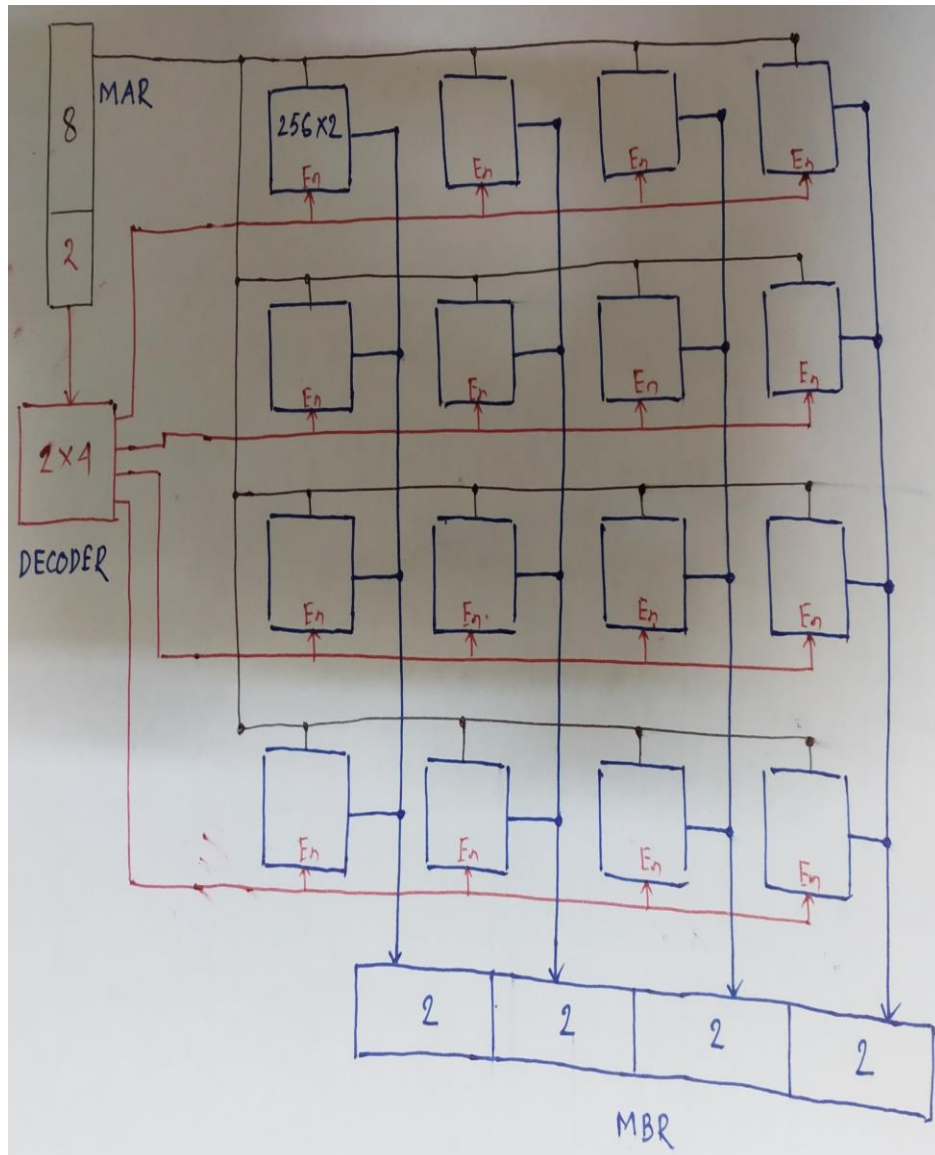


9. To design a 1024 X 8 memory using 256 X 2 chips:
- How many chips will be required? (1-mark)
- How many rows and columns will be there? (1-mark)
- Show the design layout with all the necessary units including decoder, MAR and MBR. (3-marks)

Solution:

Chips: $1024 \times 8 / 256 \times 2 = 16$

Rows: $1024 / 256 = 4$, Columns: $8 / 2 = 4$



10. Processor X is a 5-stage pipeline with a 1ns clock cycle and processor Y is a 12-stage pipeline with 0.6ns clock cycle. Due to data hazards, X experiences a stall for every 5 instructions, whereas Y experiences 3 stalls for every 8 instructions. In addition to this, the branch mis-prediction penalty for X and Y are 2 cycles and 5 cycles, respectively. Assume that branches constitute 25% of the instructions, and the mis-prediction rate for both X and Y is 5%.

- What is the average CPI of X and Y including both data hazards and branch mis-predictions?
- Based on average time to execute one instruction, which processor is faster, X or Y? Justify.

(a)

Processor X:

$$\text{\#stalls/Instruction} = (1/5) + (.25 * .05 * 2) = 0.2 + 0.025 = 0.225$$

$$\text{CPI} = 1 + \# \text{stalls/instruction} = 1 + 0.225 = 1.225$$

$$\text{Average Execution Time of one Instruction} = (1 + .225) * 1 = 1.225 \text{ ns}$$

Processor Y:

$$\# \text{stalls/Instruction} = (3/8) + (.25 * .05 * 5) = 0.375 + 0.0625 = 0.4375$$

$$\text{CPI} = 1 + \# \text{stalls/instruction} = 1 + 0.4375 = 1.4375$$

$$\text{Average Execution Time of one Instruction} = (1 + .4375) * 0.6 = 0.8625 \text{ ns}$$

(b) Processor Y is faster as the average execution time of one instruction on Processor Y is less as compared to average execution time of one instruction on processor X.

11. Assume that the original machine (P) is an 8-stage pipeline with a 1ns clock cycle. The second machine (S) is a 12-stage pipeline with 0.9ns clock cycle. The 8-stage pipeline experiences two stalls due to a data hazard for every 5 instructions, whereas the 12-stage pipeline experiences 3 stalls for every 10 instructions. In addition, branches constitute 30% of the instructions, and the mis-prediction rate for both machines is 10%. The branch mis-prediction penalty for the P and Q are is 2 and 5 cycles, respectively.

- (a) What is the average CPI of P and Q including both data hazards and branch mis-predictions?

What is the speedup of the Q pipeline over the P pipeline, taking into account both data hazard and branch mis-prediction? Justify.

Machine P:

$$\text{CPI} = \text{Ideal CPI} + (\text{Percent of branch})(\text{Branch misprediction rate})(\text{penalty}) + (\text{stall/interval})$$

$$\text{CPI} = 1 + (0.3)(0.1)(2) + (2/5)$$

$$\text{CPI} = 1.46$$

Machine Q:

$$\text{CPI} = \text{Ideal CPI} + (\text{Percent of branch})(\text{Branch misprediction rate})(\text{penalty}) + (\text{stall/interval})$$

$$\text{CPI} = 1 + (0.3)(0.1)(5) + (3/10)$$

$$\text{CPI} = 1.45$$

$$\text{a) Avg CPI of P} = 1.46 * 1.0\text{ns} = 1.460\text{ns}$$

$$\text{Avg CPI of Q} = 1.45 * 0.9\text{ns} = 1.305\text{ns}$$

$$\begin{aligned} \text{b) Speedup} &= (\text{CPI/depth})P / (\text{CPI/depth})Q \\ &= (1.46)/(1.305) \\ &= 1.1187 \end{aligned}$$

12. Consider a 5 stage MIPS pipelined processor which uses an I-cache that supply one instruction per cycle and a D-cache that is used for load-store operations. Both the caches are assumed to be perfect hit cache. Default latency of both caches is 1 cycle. The instruction mix percentage for load, branch, store and ALU operations are 30%, 20%, 8% and 42 % respectively. Out of the branch instructions two third are conditional

branch instructions and rest unconditional branch instructions. Code analysis shows that half of conditional branch instructions are always taken and rest always not taken. Generally, load instructions are offering a 2-cycle stall and branch instructions are offering 1 cycle stall. 60% of load instructions can be reordered to avoid these delay slots. 25 % of load instructions cannot be reordered. Rest of the load instructions can be reordered to fill in 1 delay slot. It was observed that 50% of conditional branch instructions and all unconditional branch instructions are schedulable to fill 1 cycle delay slots. Under these circumstances what is the resultant CPI of the machine.

(a) Without reordering / scheduling any the instructions.

(b) With reordering / scheduling the instructions for filling delay slots.

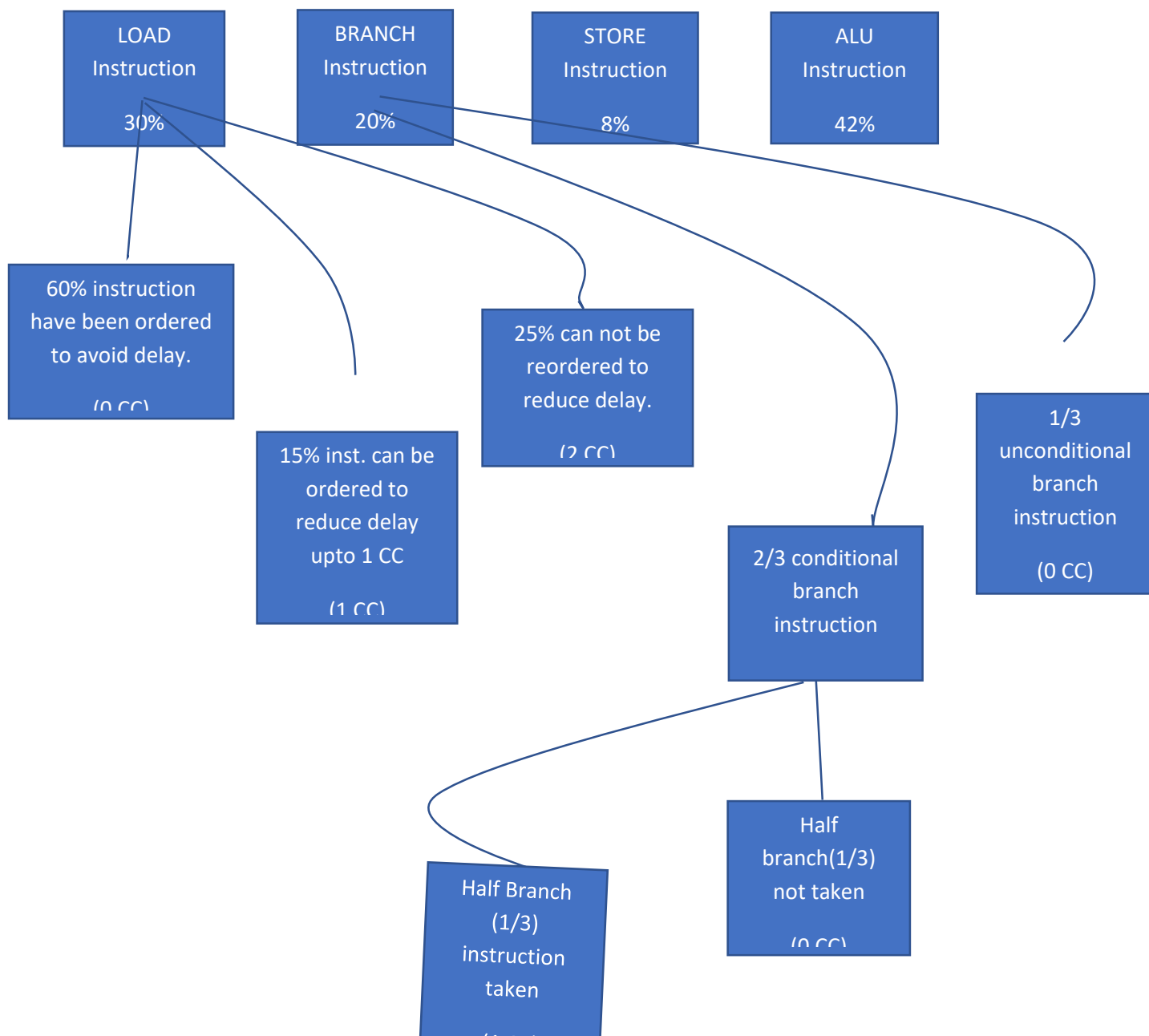
sol:

LOAD instruction -> 30%, These instructions creates 2 clock cycles STALLS

BRANCH instruction -> 20%, These instructions creates 1 clock cycle STALLS

STORE instruction -> 8%

ALU instruction -> 42%



A: No Reordering/ Rescheduling allowed :-

Total CPI = Ideal CPI + stall frequency*stall penalty

$$\begin{aligned} \Rightarrow & 1 + 0.3*2 + 0.2 * [(1/3)*1 + (1/3)*0 + (1/3)*1] + 0.8*0 + 0.42*0 \\ \Rightarrow & 1 + 0.6 + 0.2*(2/3) \\ \Rightarrow & 1.73 \text{ CPI} \end{aligned}$$

B: With Reordering / Rescheduling allowed :-

Total CPI = Ideal CPI + stall frequency*stall penalty

$$\begin{aligned} \Rightarrow & 1 + 0.3 * [0.6*0 + 0.25*2 + 0.15*1] + 0.2*[(1/3)*0 + (1/3)*1 + (1/3)*0] + \\ & 0.8*0 + 0.42*0 \\ \Rightarrow & 1 + 0.3*0.65 + 0.2*(1/3) \\ \Rightarrow & 1.2616 \text{ CPI} \end{aligned}$$

13. A microprocessor with clock frequency of 500 kHz is connected to an I/O device that employs DMA to transfer data of size 8 bytes with transmitting rate of 2KBps. During the data transfer, DMA uses 15 cycles for initialisation and 3 cycles to transfer one byte of information between memory and I/O device. Find the % time CPU is blocked in cycle stealing mode DMA.

Ans:

Transfer time = 2KBps

2 KBps in 1 sec

$$1 \text{ Bps} = 1 / 2 \times 1024$$

$$\text{For 8 bytes} = 8 / 2 \times 1024 = 3.9 \text{ ms}$$

For DMA transfer

$$f = 500 \text{ kHz}$$

$$t = 1/f = 1 / 500 \times 10^3$$

$$\text{Transfer time} = [\text{Initialization time} + \text{transfer time}] \times 8 \times \text{clock frequency}$$

$$= [15+3] \times 8 \times 1 / 500 \times 10^3$$

$$= 0.29 \text{ ms}$$

Percentage of CPU blocked = Time to transfer data /

Time required to prepare data

$$= (0.29 / 3.9) \times 100 = 7.4 \%$$

14. A microprocessor with clock frequency of 500 kHz is connected to an I/O device that employs DMA to transfer data of size 8 bytes with transmitting rate of 2KBps. During the data transfer, DMA uses 15 cycles for initialisation and 3 cycles to transfer one byte of information between memory and I/O device. Find the % time CPU is blocked in burst mode DMA.

Ans:

Transfer time = 2KBps

2 KBps in 1 sec

1 Bps = $1 / 2 \times 1024$

For 8 bytes = $8 / 2 \times 1024 = 3.9 \text{ ms}$

For DMA transfer

$f = 500 \text{ kHz}$

$t = 1/f = 1 / 500 \times 10^3$

Transfer time = [Initialization time + transfer time x 8] x clock frequency

= $[15 + 3 \times 8] \times 1 / 500 \times 10^3$

= 0.078 ms

Percentage of CPU blocked = Time to transfer data / (Time to transfer data + Time required to prepare data)

= $(0.078 / 0.078 + 3.9) \times 100 = 1.96 \%$

15. A student found a HDD inside his old PC. It was found that the disk has a rotational speed of 3600 rpm. It takes 6ms to position its head over the correct track and 1.5ms for track-to-track access. Sector size is marked as 1 KB and has 8 surfaces. Each surface has 512 tracks and each track has 64 sectors. Successive tracks in cylinder can be read without head movement. Assuming a file stored in successive sectors and tracks of successive cylinders, starting at sector 0, track 0 of cylinder i. What is the disk capacity and time required to transfer a file of size 4 MB?

Capacity = $8 \times 512 \times 64 \times 1\text{KB} = 256 \text{ MB}$

Seek time = 6ms

Size of each cylinders = 8 tracks x 64 sectors/track x 1 kB /sector = 512 KB

No of cylinders required to store 4MB file = $4\text{MB} / 512\text{KB} = 8$

Rotational latency = rotation time/2 = $60 / (3600 \times 2) = 8.3\text{ms}$

Time to find cylinder = seek time = 6ms

Time to find first sector = rotational latency = 8.3ms

Time to read all 8 tracks on one cylinder = $8 \times (60/3600) = 133.33\text{ms}$

Track to track access time = 1.5ms (given)

Time to transfer 4MB file = $6 + 7 \times (8.3+133.33+1.5) + 8.3+133.33 = 1,149.54\text{ms}$

16. A magnetic disk of size 1GB has sector size of 4KB. It has 1024 tracks per surface and 64 sectors per track. Average seek time is 8ms. Rotational disk speed is given as 3600 rpm. Considering the fact that successive tracks can be read without head movement and track to track delay is 2ms. What is the number of surfaces for the disk and also find the time taken to transfer a 5 MB file stored in successive sectors and tracks of successive cylinders from track 0, sector 0 of a cylinder?

Number of surfaces = $1\text{GB} / (4\text{KB} \times 1024 \times 64) = 4$

Seek time = 8ms

Size of each cylinders = $4 \text{ tracks} \times 64 \text{ sectors/track} \times 4 \text{ kB /sector} = 1 \text{ MB}$

No of cylinders required to store 5MB file = $5\text{MB}/1 \text{ MB} = 5$

Rotational latency = $\text{rotation time}/2 = 60 / (3600 \times 2) = 8.3\text{ms}$

Time to find cylinder = seek time = 8ms

Time to find first sector = rotational latency = 8.3ms

Time to read all 4 tracks on one cylinder = $4 \times (60/3600) = 66.66\text{ms}$

Track to track access time = 2ms (given)

Time to transfer 4MB file = $8 + 4 \times (8.3+66.66+2) + 8.3+ 66.66 = 390.84\text{ms}$

17. ABC company advertised that their magnetic disk has the following specification: Rotational speed: 3600 rpm, Average seek time: 10ms, Sector size :2KB, track to track delay 1.5ms. Number of surface: 8, tracks per surface: 256 and sectors per track: 32. Successive tracks in cylinder can be read without head movement. What is the average access time for the disk for a file stored in in successive sectors and tracks of successive cylinders, starting at sector 0, track 0 of cylinder i.? Find the time taken to transfer an 8 MB file

Seek time = 10ms

Rotational latency = $\text{rotation time}/2 = 60 / (3600 \times 2) = 8.3\text{ms}$

Average access time = seek time + rotational delay = $10+8.3 = 18.3\text{ms}$

Size of each cylinders = $8\text{tracks} \times 32 \text{ sectors/track} \times 2 \text{ KB /sector} = 512\text{KB}$

No of cylinders required to store 8MB file = $8\text{MB}/512\text{ KB} = 16$

Time to find cylinder = seek time = 10ms

Time to find first sector = rotational latency = 8.3ms

Time to read all 8 tracks on one cylinder = $8 \times (60/3600) = 133.33\text{ms}$

Track to track access time = 1.5ms (given)

Time to transfer 8MB file = $10 + 15 \times (8.3+133.33+1.5) + 8.3 + 133.33 = 2298.58\text{ms}$

18. Suppose a processor spends 75% of its time in memory operations (both I-cache and D-cache access together) when running a given program “sort” and its manufacturers make a change that improves its performance on memory access by a factor of 5. The program “sort” originally took 100s to execute.

(a) What is the speed up from the old system to the new system?

(b) What fraction of its execution time does the new system spend in memory operations?

Old Execution Time = 100 seconds

Memory access time is 75% of total time = $.75 \times 100 = 75$ seconds

Remaining Time = 25 seconds

New Execution Time:

Speed up factor in memory access time = 5

Memory access time = $75/5 = 15$ seconds

Total time = $15 + 25 = 40$ seconds

(a) Speed Up in execution time of new system = $100/40 = 2.5$

(b) Percentage of time spent in memory access = $(15/40) \times 100 = 37.5\%$

19. A program consists of two categories of instructions: S-sequential and P-parallel. The execution time for each S type or P-type instruction is 1ms on a uniprocessor. P-type instructions get speeded up by factor of N upon running on an N-core processor. The program which took 2000 ms in a uniprocessor took only 500ms upon running on a 16-core processor. Let R1 and R2 be the fraction of time spent on running S-type instruction on a uniprocessor and 16-core processor, respectively. What will be impact ratio of sequential code, if we define the impact ratio as $\frac{R1}{R2}$?

Let S be the serial part and P be the Parallel part

Unicore:

$$S+P = 2000\text{ms} \quad \dots(1)$$

16 core:

$$S+(P/16) = 500 \text{ ms} \quad \dots(2)$$

Solving (1) and (2)

$$P-(P/16) = 1500$$

$$P = 1600$$

Therefore, $S=400$ and $P=1600$

$R1$ = Time spent in running in single core

$R2$ = Time spent in running in 16 core

Unicore:

$$R1 = 400/2000$$

16 core:

$$R2 = 400/500$$

$$R1/R2 = 0.25$$

20. Given a non-pipelined architecture running at 2.5 GHz, that takes 5 cycles to finish an instruction. You want to make it pipelined with 5 stages. Due to hardware overhead the pipelined design will operate only at 'x' GHz. 30% of instructions cause a stall of 10 cycles each during the IF stage due to I-cache miss. But there is no such stall for non-pipelined architecture. Assume that there are no other hazards. It was observed that the speedup obtained by the pipelined design over the non-pipelined design is 1. What is the value of 'x'?

Execution time = CPI x clock cycle time

Speed up = Execution time unpiplined / Execution time pipelined

$$1 = \frac{5 \left(\frac{1}{2.5} \right)}{(1+0.3*10*x)}$$

Solving this $x = 2 \text{ GHz}$